Chalana Sangeeth

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Welcome

B.Sc. Engineering (Hons) candidate in **Electrical and Electronic Engineering**, having completed all academic requirements, with hands-on experience in **AI/ML, IoT and intelligent automation**. Experienced in developing practical engineering solutions involving **Generative AI, Agentic AI, Retrieval-Augmented Generation (RAG), Deep Learning, Signal Processing, and IoT**. Proficient in **Python, TensorFlow, scikit-learn, LangChain, LangGraph, and ESP32**. Available **immediately for full-time opportunities** in **AI/ML Engineering or AI/ML + IoT Engineering**.

Education

BSc (Honors) in Electrical & Electronic Engineering *Aug 2022 – Oct 2026*
South Eastern University of Sri Lanka

- GPA: 3.35/4.0

Experience

Artificial Intelligence(AI) Engineer (Intern) *Colombo, SL*
Sri Lanka Telecom *Aug 2025 – Oct 2025*

- Developed an **AI-driven back-office automation system** using **LangGraph, Gemini, RAG, and n8n** for automated request processing and information retrieval.
- Built **agentic AI workflows and Flask REST APIs** for usage analysis and document-based problem resolution, with **LangSmith** for workflow monitoring and debugging.

Software Engineer (Intern) *Colombo, SL*
Sri Lanka Telecom *May 2024 – Aug 2024*

- Developed and enhanced **PowerProx**, an internal asset-management system for tracking **rectifiers, generators, batteries, SPDs, and UPS systems**.
- Built **Flutter/Dart features and REST API integrations**, digitizing asset management and reducing manual record handling.

Projects

Elephant Detection Using Seismic Signals *Ongoing*

- Developing an **AI/ML + IoT system** for elephant detection using **geophone-based seismic signals** and **signal processing**.
- Implementing **ML classification** and **real-time embedded AI** using **ESP32/ESP32-S3**.

Industrial Engineering Project – PowerProx (SLT) May 2024 - Aug 2024

- Developed **Flutter/Dart** features and integrated **REST APIs** for real-time asset tracking and updates.
- Integrated **Gemini API** for **AI-assisted quotation retrieval and processing**.
- **Tools:** Flutter, Dart, REST APIs, Gemini API, **Generative AI**

Smart Route Optimization System *May 2025 -May 2025*

- Developed a **Flask-based** route optimization system using **Dijkstra's algorithm** and **Google Maps API**.

Accommodation Booking Mobile App *Apr 2025 - Jan 2026*

- Developed a **Flutter** application with **Firebase authentication** and image uploads.
- Built **Flask REST APIs** with **MongoDB** for property management.

Certifications

Advanced Learning Algorithms *Jun - 2026*
- Issued by: DeepLearning.AI on Coursera

AI / ML Engineer - Stage 1 *Apr - 2025*
- Issued by: Centre of Open and Distance Education - SLIIT

AI / ML Engineer - Stage 2 *Apr - 2025*
- Issued by: Centre of Open and Distance Education - SLIIT

Supervised Machine Learning - Regression and Classification *Jul - 2024*
- Issued by: DeepLearning.AI on Coursera

AgniXtream Hackathon Competition *Dec - 2024*
- 1 Runners Up at 6 hours Hackathon
- Issued by: IEEE Sri Lanka Section

Technologies

Programming: Python, C++, Dart

AI / Machine Learning: Machine Learning, Deep Learning, TensorFlow, scikit-learn, NumPy, Pandas

Generative & Agentic AI: RAG, LangChain, LangGraph, Gemini API, Agentic AI, n8n, LangSmith

Signal Processing: Digital Signal Processing, Seismic Signal Processing, Filtering, FFT, Feature Extraction

Embedded & IoT: ESP32, ESP32-S3, Embedded AI, Sensors, ADC, IoT Systems, Real-Time Data Acquisition

Backend & Software: Flask, REST APIs, MongoDB, Firebase, Flutter, Git

Tools: MATLAB

References

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